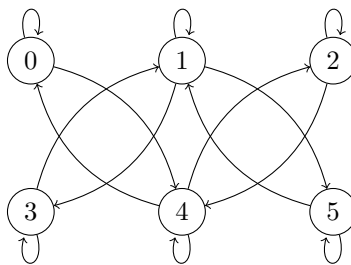


Problem 1 (§11.1, #4). Here is a diagram for a relation R on a set A . Write the sets A and R .



Problem 2 (§11.2, #14). Suppose R is a symmetric and transitive relation on a set A , and there is an element $a \in A$ for which $a R x$ for every $x \in A$. Prove that R is reflexive.

Problem 3 (§11.2, #16). Define a relation R on $\mathbb{Z}$ by declaring that $x R y$ if and only if $x^2 \equiv y^2 \pmod{4}$. Prove that

- a) R is reflexive.
- b) R is symmetric.
- c) R is transitive.

Problem 4 (§11.3, #8). Define a relation R on $\mathbb{Z}$ as $x R y$ if and only if $x^2 + y^2$ is even.

- a) Prove that R is an equivalence relation.
- b) Describe its equivalence classes.

Problem 5 (§11.4, #6). Consider the partition $P = \{\{0\}, \{-1, 1\}, \{-2, 2\}, \{-3, 3\}, \{-4, 4\}, \dots\}$ of $\mathbb{Z}$. Describe the equivalence relation whose equivalence classes are the elements of P .

Problem 6 (§11.5, #5 (adapted)). Suppose p is a prime number and let $[a], [b] \in \mathbb{Z}_p$. Prove or disprove the statement below.

$$\text{If } [a] \cdot [b] = [0], \text{ then } [a] = [0] \text{ or } [b] = [0].$$

Problem 7 (§11.5, #6 (adapted)). Suppose p is a prime number and let $[a], [b] \in \mathbb{Z}_{p^2}$. Prove or disprove the statement below.

$$\text{If } [a] \cdot [b] = [0], \text{ then } [a] = [0] \text{ or } [b] = [0].$$

Problem 8. Recall that for a complex number $x = a + bi$, we define $|x| = \sqrt{a^2 + b^2}$. Define a relation R on $\mathbb{C}$, the set of complex numbers, by $x R y$ whenever $|x| = |y|$.

- a) Show that R is an equivalence relation.
- b) Describe the partition of $\mathbb{C}$ induced by R .